

PART A

- 9 Define Sub threshold conduction in MOSFET (3)
- 10 Define threshold Voltage of MOSFET. How it can be varied? (3)

9)

For the conduction to happen in the MOSFET; we need the V_G to be greater than the threshold voltage V_T .

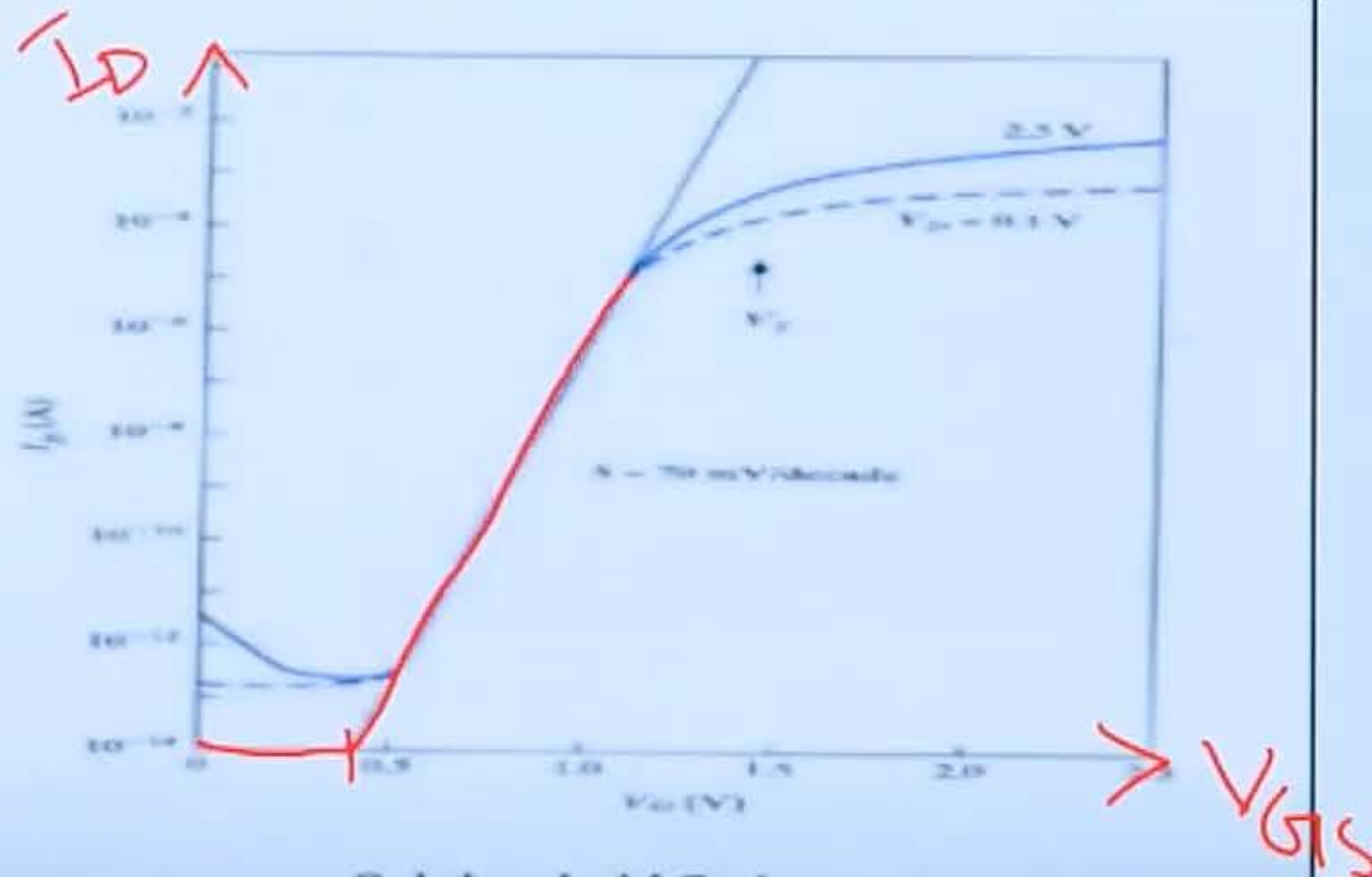
But, this threshold voltage is calculated at the point where the region below the oxides has entered into strong inversion.

$$V_G < V_T$$

From experimental results, one can observe that there is still some non-zero current flowing from drain to source even when we are operating at a region with $V_G < V_T$ (sub-threshold region).

This happens because, for the **subthreshold region**, the substrate near oxide-interface is in "**WeakInversion**". At this point, if we apply a positive V_{DS} , there will be a small current I_D flowing. We have

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_t) V_{DS} - \frac{V_{DS}^2}{2} \right]$$



10)

It is the voltage required to introduce strong inversion in the semiconductor.

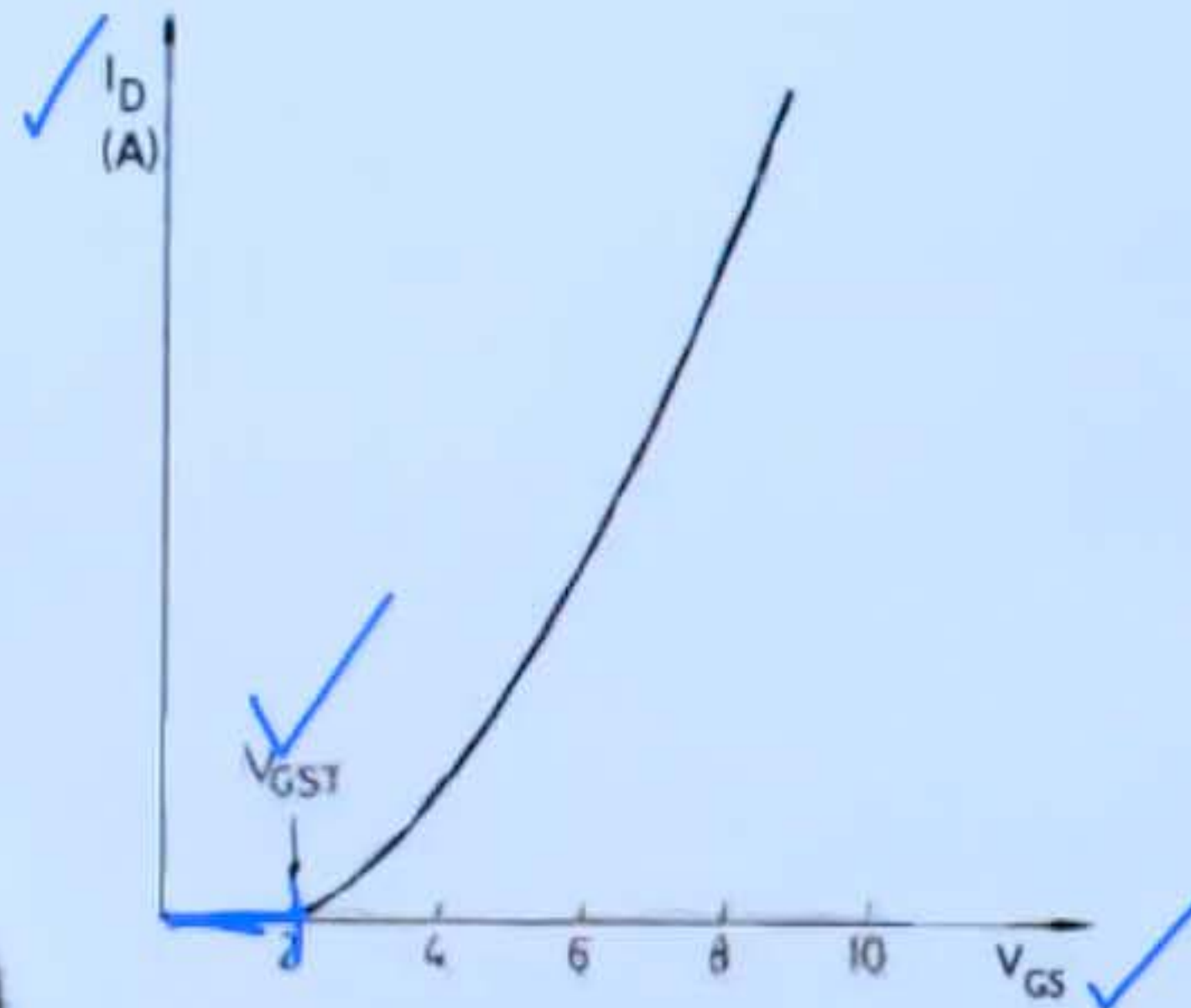
 V_T 

Fig.b Transfer Characterstic

$$V_{TN} = V_{TO} + \gamma \left[\sqrt{2|\phi_F| + V_{SB}} - \sqrt{2\phi_F} \right]$$

$$V_{TO} = \underbrace{V_{FB}} + \underbrace{2\phi_F} + \frac{\sqrt{2\epsilon_{Si}qN_A(2\phi_F)}}{C_{ox}}$$

$$\underline{\underline{V_{FB} = \left(\phi_{ms} - \frac{Q_{ss}}{C_{ox}} \right)}}$$

$$\checkmark \phi_F = \frac{kT}{q} \ln \left(\frac{N_A}{n_i} \right)$$

$$C_{ox} = \frac{\epsilon_{ox} A}{W}$$

PART B

- 19 (a) What is drain induced barrier lowering. Discuss its effect on MOSFET (6)
performance.
- (b) What is MOSFET scaling? Explain different types of scaling. Discuss the (8)
advantage and disadvantage of scaling.

19 a)

Short Channel Effects

- Short-channel effects occur in MOSFETs in which the channel length is comparable to the depletion layer widths of the source and drain junctions.
- A MOSFET device is considered to be short channel device when the channel length is the same order of magnitude as the depletion-layer widths (x_{dD} , x_{dS}) of the source and drain junction. (That is, the effective channel length L_{eff} is approximately equal to the source and drain junction depth x).

- Different short-channel effects include
 1. Channel Length Modulation
 2. Drain-induced barrier lowering and "Punch through"
 3. Velocity saturation
 4. Threshold voltage variations
 5. Hot carrier effects

Drain Induced Barrier Lowering (DIBL)

- When the depletion regions surrounding the drain extends to the source, so that the two-depletion layer merge (i.e., when $x_{dS} + x_{dD} = L$), punch through occurs.
- Punch through can be minimized with thinner oxides, larger substrate doping, shallower junctions, and obviously with longer channels.
- The current flow in the channel depends on creating and sustaining an inversion layer on the surface.
- If the gate bias voltage is not sufficient to invert the surface ($V_{GS} < V_T$), the carriers (electrons) in the channel face a potential barrier that blocks the flow. Increasing the gate voltage reduces this potential barrier and, eventually, allows the flow of carriers under the influence of the channel electric field.

- In small-geometry MOSFETs, the potential barrier is controlled by both the gate-to-source voltage V_{GS} and the drain-to-source voltage V_{DS} .
- If the drain voltage is increased, the potential barrier in the channel decreases, leading to drain induced barrier lowering (DIBL). The reduction of the potential barrier eventually allows electron flow between the source and the drain, even if the gate-to-source voltage is lower than the threshold voltage. The channel current that flows under this condition ($V_{GS} < V_T$) is called the sub-threshold current

19 b)

MOSFET SCALING

One approach to size reduction is a scaling of the MOSFET that requires all device dimensions to reduce proportionally.

The main device dimensions are the channel length, channel width, and oxide thickness. Lateral dimensions such as channel length and width are reduced by a factor of k , so should the vertical dimensions such as source/drain junction depths and gate insulator thickness.

Scaling improves

- ✓ 1. Packing density
- ✓ 2. Speed
- ✓ 3. Power dissipation

The design of high density chip in MOS VLSI technology requires high packing density of MOSFET ie; the size of transistors are as small as possible

Two types of scaling are common:

- (i) constant field scaling
- (ii) constant voltage scaling

Full scaling (constant-field scaling) –

- All dimensions are scaled by k and the supply voltage and other voltages are so scaled
- Magnitude of internal electric field is kept constant
- Only lateral dimensions are changed
- Threshold voltage is also affected

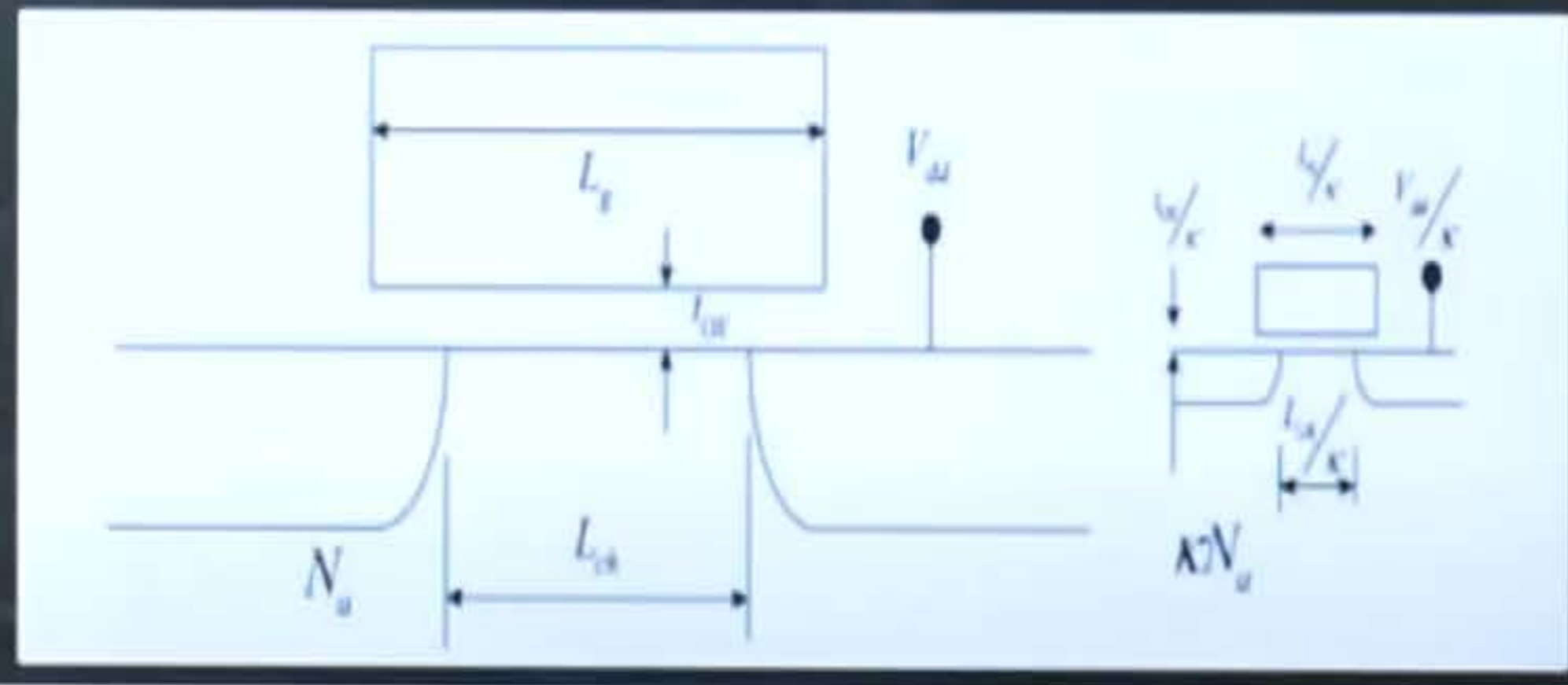


Table 6-1 Scaling rules for MOSFETs according to a constant factor K . The horizontal and vertical dimensions are scaled by the same factor. The voltages are also scaled to keep the internal electric fields more or less constant, and the hot carrier effects manageable.

	Scaling factor
Surface dimensions (L, Z)	$1/K$
Vertical dimensions (d, x_i)	$1/K$
Impurity Concentrations	K
Current, Voltages	$1/K$
Current Density	K
Capacitance (per unit area)	K
Transconductance	1
Circuit Delay Time	$1/K$
Power Dissipation	$1/K^2$
Power Density	1
Power-Delay Product	$1/K^3$

Disadvantages - Hot electron effects & Short channel effects

Constant-voltage scaling

The voltages are not scaled and, in some cases, dimensions associated with voltage are not scaled

Parameter	Scaled parameter
Channel length (L)	$1/\alpha$
Junction depth (x_j)	$1/\alpha$
Substrate doping (N_A)	α
Depletion layer thickness (d)	$1/\alpha$
Transconductance (g_m)	α
Static power dissipation (P_{static})	α
Dynamic power dissipation (P_{dyn})	α
Current (I)	α
Gate delay (τ_p)	$1/\alpha^2$
Load capacitance (C_L)	$1/\alpha$
Channel width (W)	$1/\alpha$
Supply voltage (V)	1
Gate oxide thickness (t_{ox})	$1/\alpha$

Comparison

Quantity	Sensitivity	Constant Field	Constant Voltage
Scaling Parameters			
Length	L	$1/S$	$1/S$
Width	W	$1/S$	$1/S$
Gate Oxide Thickness	t_{ox}	$1/S$	$1/S$
Supply Voltage	V_{dd}	$1/S$	1
Threshold Voltage	V_{th}	$1/S$	1
Doping Density	N_A, N_D	S	S^2

Constant voltage scaling is widely used.



PART B

- 20 (a) Explain different types of short channel effects in MOSFET. (8)
- (b) Draw and explain the structure and working of Fin FET. (6)
-



20 a)

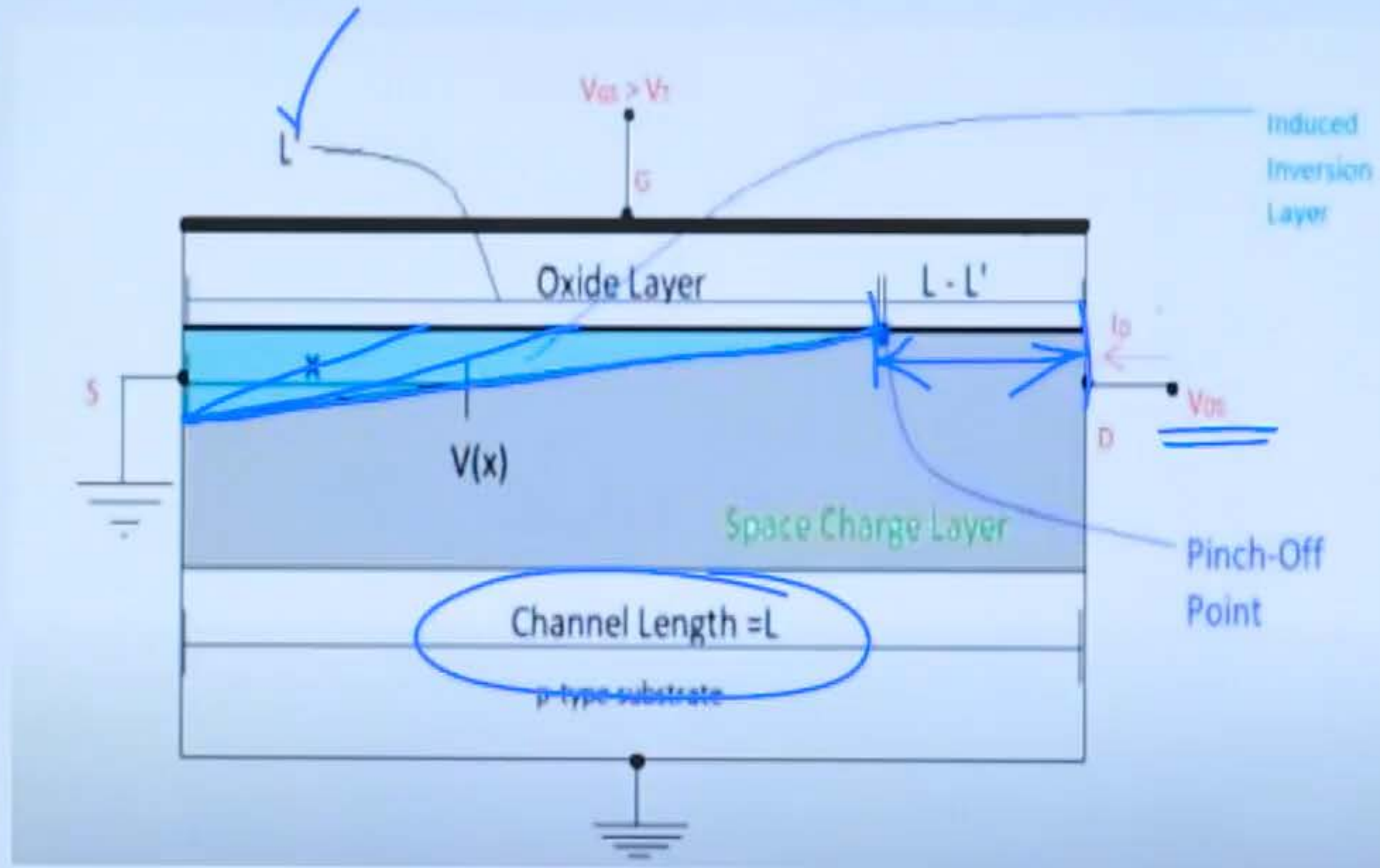
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Channel Length Modulation(CLM)

- As we keep on increasing V_{DS} , the region for which the inversion charge is zero keeps on increasing for a constant value of V_{GS} maintained. Thus channel length keeps on decreasing. This phenomenon is called Channel Length Modulation.
- This is similar to "Base Width Modulation"
- Thus we get a V_{DS} term in the expression for I_D even when we are operating in the saturation region. Generally, the fabrication of the MOSFET devices is done in a way such that the change in length given by $\Delta L = L - L'$ is low with a change in V_{DS} .

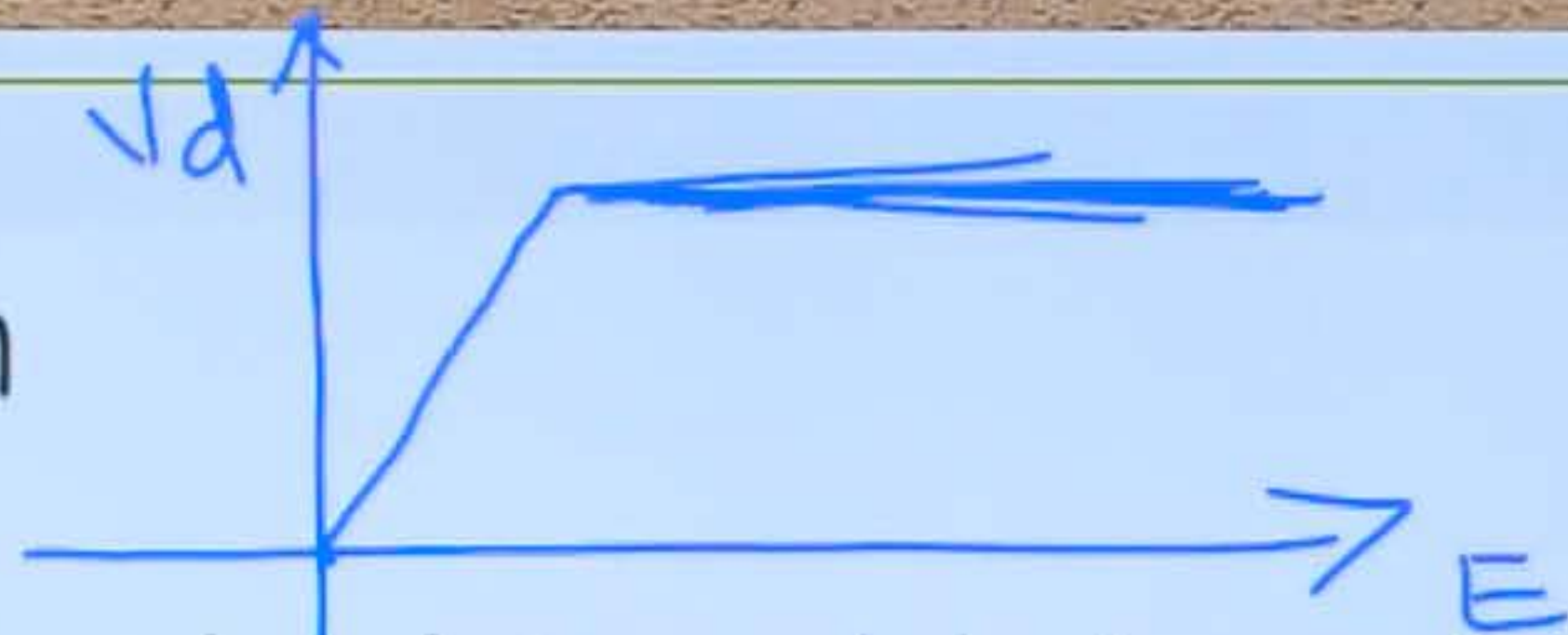


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Velocity Saturation



- The velocity of charge carriers, such as electrons or holes, is proportional to the electric field that drives them, but that is only valid for small fields.
- As the field gets stronger, their velocity tends to saturate.

$$V_d \propto E$$

$$V_d = \mu_n E$$

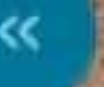
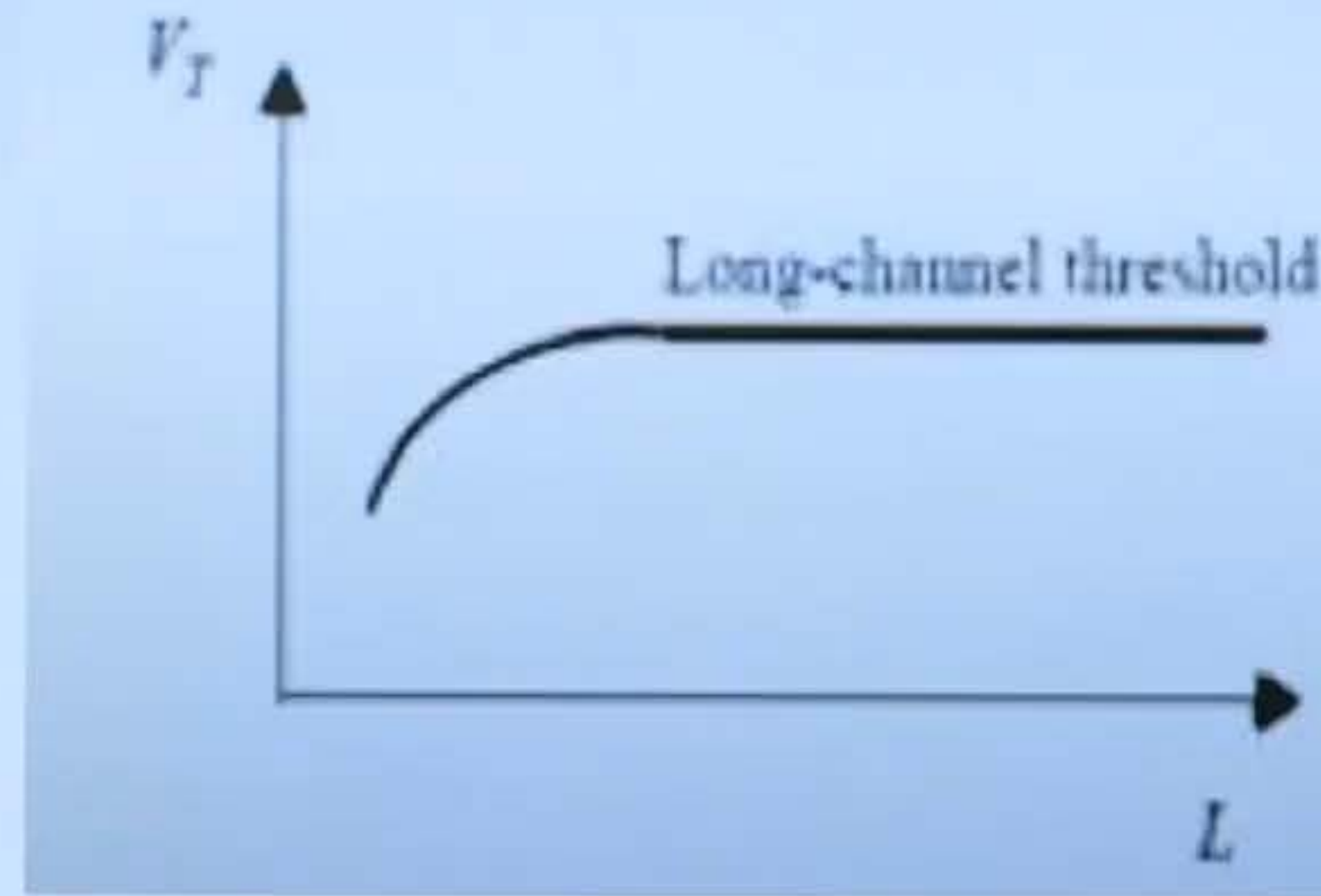
- That means that above a critical electric field, they tend to stabilize their speed and eventually cannot move faster.

- Velocity saturation is specially seen in short-channel MOSFET transistors, because they have higher electric fields

- The drift velocity of the electrons in the inversion layer to be proportional to the lateral electric field applied. The proportionality constant was given by μ_n .

Threshold Variations

- The threshold voltage is only a function of the manufacturing technology and the applied body bias V_{SB} .
- The threshold can therefore be considered as a constant over all NMOS (PMOS) transistors in a design. '
- As the device dimensions are reduced, this model becomes inaccurate, and the threshold potential becomes a function of L , W , and V_{DS} .
- Two-dimensional second-order effects that were ignorable for long-channel devices suddenly become significant.
- In the traditional derivation of the VTO, for instance, it is assumed that the channel depletion region is solely due to the applied gate voltage and that all depletion charge beneath the gate originates from the MOS field effects.



PART B

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-



20 b)

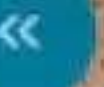
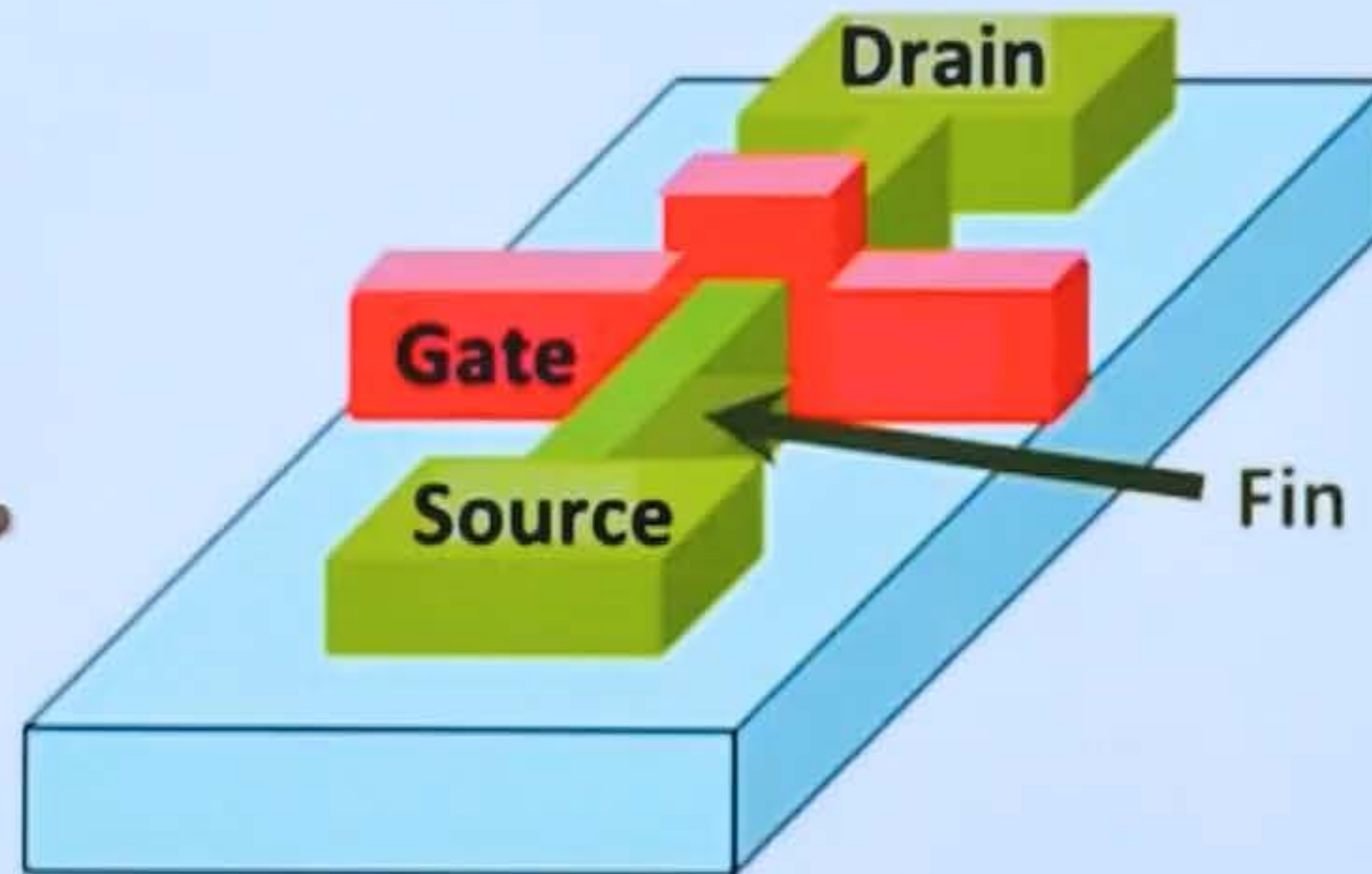
FinFET

- FinFET, also known as Fin Field Effect Transistor
 - It is a type of non-planar or "3D" transistor used in the design of modern processors
 - FinFETs are new generation transistors which utilize tri-gate structure (Multi gate device).
 - It is used more than other FETs because of its area of performance, lower leakage power, low voltage operation, intra-die variability and lower retention voltages for SRAM.
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- The conducting channel is greatly controlled by the gate.
 - The thickness of the fin (measured in the direction from source to drain) determines the effective channel length of the device.

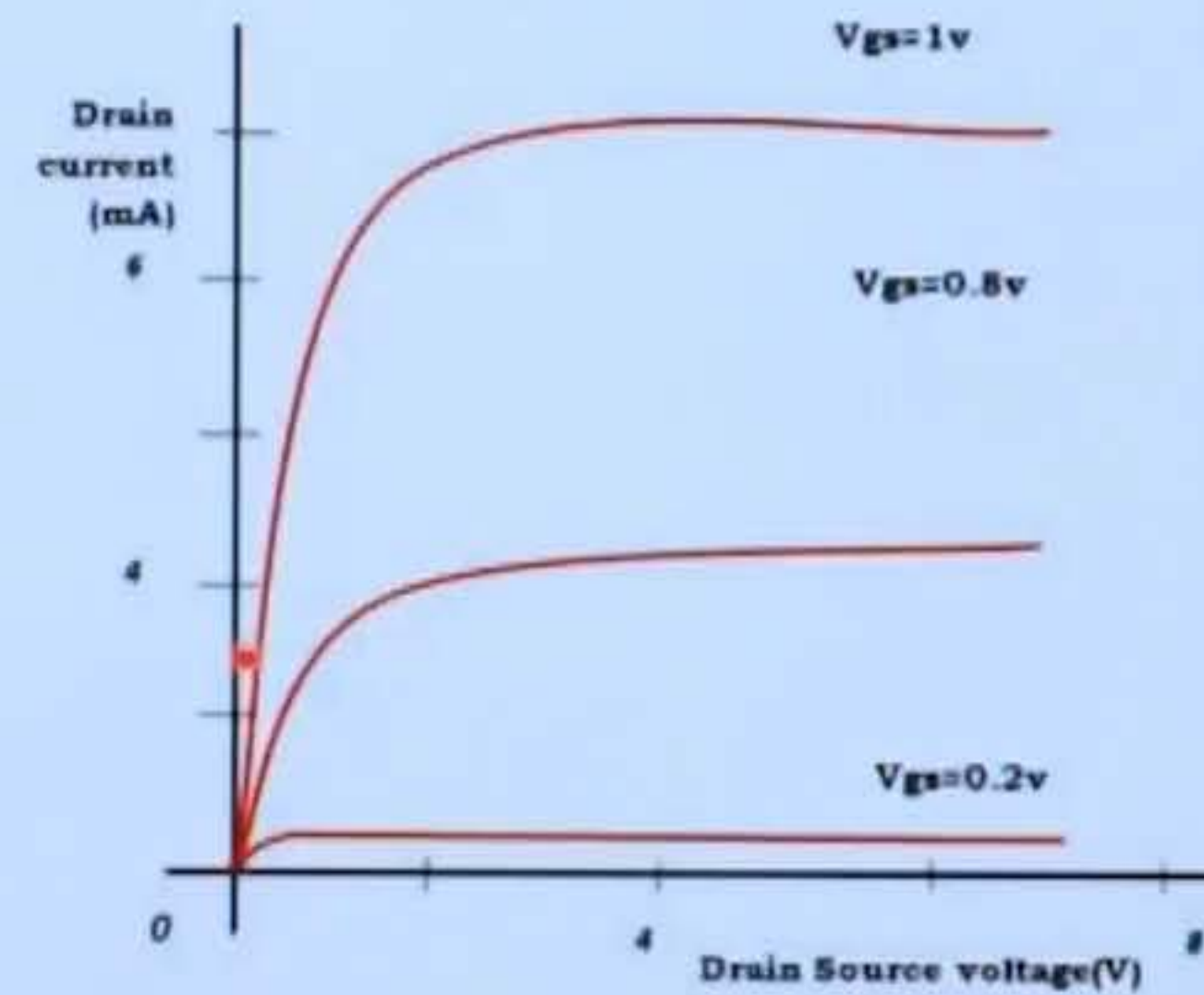
Advantages

- Reduced short-channel effects (SCEs) and leakage current .
- To overcome the worst types of short-channel effect encountered by deep submicron transistors, such as drain-induced barrier lowering (DIBL).
- This technique provides increased operating speed by low-threshold MOSFET and reduced leakage by high-threshold voltage.
- The wrap around gate structure provides a better electrical contact over the channel and thus helps in **reducing the leakage current** and overcoming the short channel effects.
- Higher drain current
- Lower switching voltage
- ~~Low power consumption~~

STRUCTURE



V-I CHARACTERISTICS OF FINFET



Transfer characteristics of FinFET

